

ITS-U: Intelligent Tracking System for You

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ABSTRACT

Monitoring athletes' positions during outside events has always been a demanding task. Current tracking systems might not suit the needs of certain athletic events, especially those in harsh and rough conditions. This paper pushes forward ITS-U, an innovative RFID-based timing system that implements IoT paradigms to facilitate the tracking of runners during athletic or trekking events. The solution intends to provide the necessary data gathering information stations and equipment to record athletes' information during the route and pass this on to organizers and emergency services when required.

Keywords

Radio Frequency Identification, Internet of Things, Real-time monitoring, Tracking, Timing system

1. INTRODUCTION

1.1 Problem Definition

Physical exercise in the form of running, jogging, or walking is regarded as one of the healthiest ways of exercising and keeping fit [28]. This exercise can be practiced leisurely by way of personal fitness or hiking, as well as competitively in events such as marathons or jogging competitions. Over the years, events have become more demanding and competitive [11], and so, the need to detect and track runners in mass races or even sporting events such as cycling competitions and hiking has rapidly increased.

Tracking systems for sports events frequently make use of Radio Frequency Identification (RFID) as shown in [20]. With the rapid increase of Internet of Things (IOT) devices, RFID technology can be used to gather substantial amounts of data from monitoring and locating of participants [19]. By using an RFID reader, an RFID antenna, and a number of RFID tags, the time taken to register all of the participants is massively reduced and the number of errors is decreased. Such a system would be able to track the starting time of the participants, together with the timing of the participants through each checkpoint, as well as the finishing time. As such, the timing of each participant can be displayed in real-time. In cases of extreme sports, the timing of the participants can be passed on to emergency services when required or used accordingly when an emergency arises.

1.2 Motivation

Organizing and surveilling outdoor running and trekking

events brings with it a number of complications. Enrolling the required workforce during the registration of the participants and the observation and verification of the contestants at the starting and finishing checkpoints is time-consuming and could easily lead to errors [14]. Moreover, the staff assigned to handle and track the runners tend to make mistakes when recording timestamps and keeping records. The use of manual timing systems such as stopwatches is not viable, and considering that the difference between runners could be up to a tenth of a second, the final results of a racing contest could be heavily affected [11].

2. AIMS AND OBJECTIVES

This project goes out to investigate the automatic tracking of persons in the outdoors during sports events while employing technologies to accurately determine their location but also optimise their performance through the collected analytics.

A number of measurable objectives have been set to achieve the aim stated above, namely:

- **Objective (O1):** To investigate and employ tracking techniques through wearable devices while considering state-of-the-art technologies;
- **Objective (O2):** To research, adopt and develop intelligent techniques to optimise the use of collected tracking data in real time;
- **Objective (O3):** To design, develop and test a prototype system to test and evaluate the proposed intelligent tracking system.

3. BACKGROUND AND LITERATURE REVIEW

In this chapter, the aim is to discuss existing monitoring systems and techniques used in athlete tracking. In the first part, we evaluate the key components that comprise RFID technology and how the IoT paradigm is incorporated. We then examine state-of-the-art RFID systems. Lastly, we explore the retrieval of data and efficient tagging through RFID Tagging on users.

3.1 RFID Technology

RFID is an electronic wireless technology that utilizes radio waves [2, 31] to perform localization on humans and objects in motion [30]. Using RFID leads to automatic data retrieval without any need for human intervention in the

process [40, 43]. An RFID system can detect numerous objects including humans, vehicles, cargo, freight, and assets [26], up to a certain distance that are not in the line of sight [5]. A typical RFID system is made up of an RFID reader, an RFID antenna, and a number of RFID tags, as shown in Figure 1.

Since employing such a system is economically feasible, long-lasting, and energy-efficient [44], it is extensively used in different IoT applications to improve their benefits [1]. RFID has appealed to a multitude of research endeavors due to its inventive capability in a diversity of applications [7]. Such applications include healthcare and hospital information systems [27, 39], traffic monitoring [2], baggage tracking [33], payment systems [32], animal tracking [3], and anti-theft systems [22].

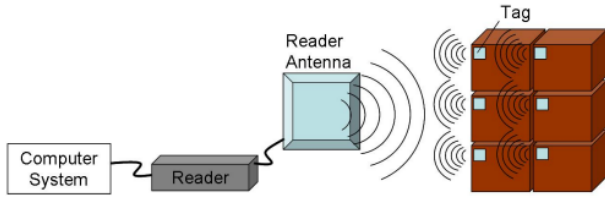


Figure 1: Configuration of an RFID System [36]

3.1.1 RFID Tags

An RFID tag (also known as an RFID transponder) [38, 42] is a wearable device that comes in a variety of shapes, sizes, and capabilities, as shown in Figure 2. It consists of 3 main components including an antenna, a silicon chip, and a substrate that holds together the antenna and the chip [37]. All of these components combined are called an RFID inlay. Three types of tags exist, including passive, semi-passive, which are also named battery-assisted passive (BAPs) [21], and active. The main difference comes from the assembly of the components found inside these tags. The type of tag that is used relies upon the business type and technological necessities of the company [17].



Figure 2: Types of RFID Tags [15]

Passive tags cannot power themselves since they do not contain a battery. Table 1 displays the different frequency ranges that a passive tag can have, together with its read range and tag reading. The frequency used is based on the tag's application [6]. Passive tags gain power with the use of the RFID reader. On the other hand, BAPs and active tags can power themselves since they have an internal battery.

Band	Low Frequency	High Frequency/ NFC	Ultra High Frequency
Frequency	125-134 kHz	13.56 MHz	865-956 MHz
Read Range	Less than 10cm	Up to 1m	Up to 12m
Tag Reading	Single reads	Multiple	Excellent Multiple

Table 1: Different RFID Frequencies [9]

RFID tags are assembled to abide by a categorization of classes. In their work, Jia et al. [42] illustrate six types of classes for RFID tags, starting from Class 0 up to Class 5. Class 0 tags are factory programmed, while Class 1 tags are user adjustable. Both Class 0 and 1 provide basic RF potential. Class 2 tags allow read and write RF memory together with the possibility of encryption. Class 3 tags implement battery powering as well as further read range and broadband communication. Class 4 tags include further sensing and peer-to-peer (P2P) communication. Class 5 tags can be considered to be a reader since they have the capability to trigger other tags with the power they contain. Passive tags fall under the Class range of 0 to 3, with BAPs falling under Class 3. On the other hand, active tags fall under Class 4, while active tags that are able to gather data from other tags and tag readers fall under Class 5.

3.1.2 RFID Antenna

An RFID antenna is used to receive and transmit electromagnetic waves so as to recognize RFID tags and communicate data from the tags to the RFID reader. A multitude of antennas exist, such as gate antennas, patch antennas, or even mats. The performance of antennas depends on a number of aspects, including a satisfactory data transfer rate, a large radiation efficiency, robustness, as well as having a small side-lobe to decrease interference [18, 37]. Together, a tag and an antenna make up the wireless communication interface protocol [10].

3.1.3 RFID Reader

An RFID Reader (also known as RFID interrogator [38] or transceiver [42]) is used to read and write data with compatible RFID tags. A reader is made up of three main components including a high frequency interface, a control section, as well as an in-built antenna [6]. It transmits energy to an antenna, which transforms that energy into a Radio Frequency (RF) wave. Once the tag picks up the RF wave, the antenna inside the tag moves the energy to the silicon chip. The chip sends a signal back to the reader that includes a unique digital ID (identifier) [31]. This process is called backscatter. Data can be passed over lengthy distances, even in extreme environmental conditions such as dirt, dust, oil, and even paint [8].

3.2 Internet of Things

By combining RFID together with the terminal of Internet, the readers are able to establish, monitor, and record objects that are attached with tags systematically in real-time. Altogether, this is the IoT, which connects physical and virtual objects together with the capability of communication and data retrieval [42]. As such, RFID is considered one of the main pivotal enablers of the IoT [13, 29, 34]. RFID tags, which are considered to be IoT-enabled wearables, can be implanted or attached to clothing and apparel, or even stickered onto objects [12].

In Figure 3, the development of IoT is seen to correspond with that of RFID technology. The National Informatics Centre (NIC) anticipates that "by 2025, Internet nodes may reside in everyday things – food packages, furniture, paper documents, and more" [4].

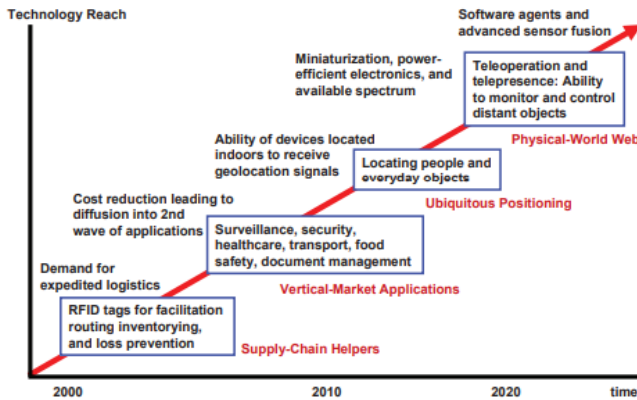


Figure 3: Internet of Things Roadmap [42]

3.3 RFID Tracking Systems

3.3.1 Running and Jogging Events

In 2015, a paper published by Joolrat et al. [16] discussed their real-time processing marathon system that makes use of ultra-high frequency (UHF) RFID technology for the Khon Kaen International marathon race. This system implements four checkpoints that track the time that the participants clock at each one. These include a check-in point, a start point, checkpoint stations along the track, and a finish point. In every checkpoint, a control station is installed, together with an RFID antenna that is positioned under plastic panels on the floor. Each participant had an RFID tag attached to both shoes under their shoelaces, which enabled the antenna to detect each participant and display a real-time result on a monitor. Uploading the final results to an online web application allowed participants to view the standings of the race in real-time. Such a system decreased the processing time of the updating of marathon results from one hour to five minutes, and also lowered the price to about \$5,714 from the estimated \$57,142 of a fully integrated commercial system.

However, investing in an already existing RFID timing system or the system mentioned in [16] might not be economically feasible to be implemented in smaller scale events. To find a solution for this, Satitsuksanoh et al. [28] proposed an economical interactive platform that utilizes low-frequency RFID technology. Since low-frequency RFID has

a short reading range of up to 10cm [9], the system positioned the RFID reader at waist level to negate the problem. Three different RFID tags were tested randomly between 100 users to attain the optimal way of tracking the runners. The tags consisted of a silicon wristband, a normal wristband with an exposed RFID tag, or a keychain. The worst performing tag was the silicon wristband due to its thicker casing, while the wristband with the exposed RFID tag performed the best with a reasonable recognition time. With the use of an online database, the handling of data was done by displaying the runners' information including their timing and laps covered in real-time.

Optimizing RFID systems in order to identify participants in large events is not a straightforward task. As a result of this, Polivka et al. [25] worked on a system study that analyzed the fundamental causes of failures in past RFID systems. The enhanced RFID system was analysed in a common outdoor activity, and concentrated on the improvement of more effective, well-positioned and properly aligned antennas for both the RFID reader and RFID tags. Seven participants in an in-line structure tested the system with two speeds. The first speed was fast walking at around 6km/h, while the second speed was running, at around 15km/h. The final results showed that with the standard RFID system, the identification rate of walking and running combined was 59.6%, while the enhanced RFID system had an identification rate of 100%.

3.3.2 Hiking

Hiking is considered to be an extreme sport that allows people to lead an active lifestyle. However, such a sport carries a substantial risk with it. Research shows that from 1996 till 2014, a staggering 134 incidents occurred in the Malaysian mountains, with 52 of them involving missing people [23].

In their work, Wu et al. [41] put forward a system for recreational hiking, where RFID tags are implanted in trees inside a forest. An RFID reader is provided to the hikers, with the possibility of having it implemented into their mobile phones. Hikers are able to read data from the tags which would aid them while maneuvering through the forest, and also write data to the tags in the form of an identifier as well as Sequence Numbers (SNs) which increment with each tag, forming a virtual path that can be backtracked. Such a system would be able to assist emergency services when a search and rescue operation is required by allowing the emergency responders to follow the path that the hiker had left.

In relation to the work in [41], Masohor et al. [23] introduced a system in which users are provided with an RFID tag that is used to check in at checkpoints positioned along a mountain track. The checkpoints are able to store the hikers' data with the use of a memory card, which in turn eases the work required when a search and rescue operation is needed. Emergency services are able to determine the final checkpoint that the hiker checked in and reduce the search area, which massively reduces the search time. Apart from this, this system is also able to deliver a notification message with a date, time, and the user that checked in through the use of Telegram. This feature applies solely to the first checkpoint, however, further implementation could lead to all checkpoints having this feature.

3.3.3 Cycling Events

Cycling has become an occasional sport nowadays, alongside famous bicycle races being held all over the world with the likes of Tour de France, Giro d'Italia in Italy, as well as the Vuelta de España in Spain. Nonetheless, the tracking of cyclists during these events is no less important in order to output the correct results in real-time.

In order to cater for such events, Misran et al. [24] work on a Smart Detector Database using RFID (SDDR) to monitor cyclists throughout Le Tour De Langkawi, a Malaysian bicycle race. Such a race would require a numerous amount of people to be able to track the racers smoothly. In order to avoid this, the system proposed utilizes UHF to monitor cyclists at each checkpoint, who are given an adhesive passive RFID tag each that is stickered on the bicycle. The data gathered includes the date, arrival time of each checkpoint, and the cyclists' personal information. This is then forwarded to the organizers at the last checkpoint of the race. This system managed to reduce the amount of staff required, as well as having more accurate results published in real-time.

Moreover, Tsai et al. [36] discuss and prototype a bicycle race monitoring system using passive RFID tagging together with UHF RFID technology. The aim of the system was to precisely monitor and record all of the racers in an adequately rapid manner, by minimally affecting the race while keeping the overall cost of the system as low as possible. The system was composed of a truss antenna deployment together with the a four novel RFID antenna deployment. Since bikes are made from metal and/or graphite fibers, these would affect the RF of passive tags, which led to the decision that the RFID tags would be stuck on the helmets, as can be seen in Figure 4. The results showed that the RFID tag reading was nearly 100% effective for this solution.

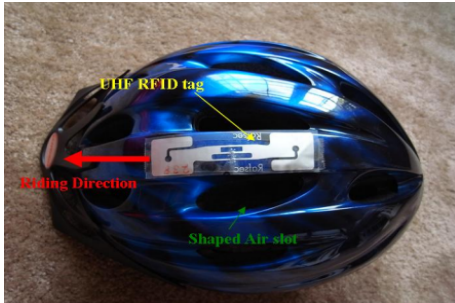


Figure 4: Helmet Tagging [36]

3.4 RFID Tagging

The placement of RFID tags is an essential factor in sports event monitoring since it influences the type of tag and hardware required to accurately monitor and track participants, as well as the outcome and results of such events. RFID tagging methods all have their strengths and weaknesses depending on their use. The most popular types of RFID tagging are shown in Figures 5 and 6. In their race timing system, Tengrunroj et al. [35] make use of shoe tagging to achieve a 100% read rate in their race events.

In the performance analysis study performed in [14], the system put forward makes use of two panel antennas to-

gether with an RFID reader, which is adaptable for a group of 50 to 200 participants. Having a read range of 1.83m is necessary to assure that the RFID tags are read successfully. 100 tests were performed to find the optimal way of tracking participants. In the case of RFID tagging, tests included bib tagging, double bib tagging, shoe tagging, as well as hip tagging. Apart from the latter, tests were held to establish the optimal positioning of antennas. This was done by experimenting with different heights. The analysis made confirmed that the ideal setup would have the tags 75cm above the ground, while employing the antennas at floor level.

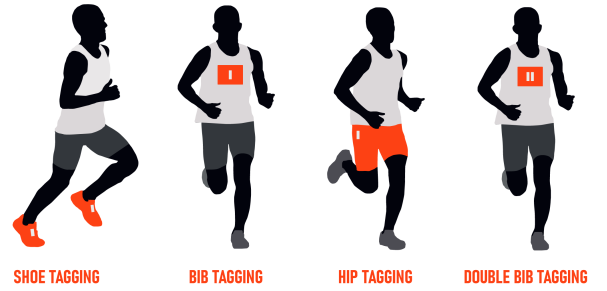


Figure 5: Different RFID Tagging [14]



Figure 6: Wrist and Ankle Tagging [14]

4. PROPOSED SOLUTION

The applications which have adopted RFID monitoring systems have been analyzed in the preceding sections. Employing this tracking system in sport events which are located in rugged environments would aid the event organizers to better plan out and maintain the event, while also aiding emergency services whenever an incident arises. As such, this research will implement a RFID system which can be widely used in different environments while employing IoT technology to further enhance the results of this system in real-time.

The system would need to be tested to allow the RFID system to detect multiple tags simultaneously, since in a real world environment, it is likely that there will be several instances where more than one athlete or object passes by the RFID reader at a given time. Before anything else, the system would need to be designed and adapted in a way that it is able to detect humans or objects at a certain range.

In order to precisely monitor every single racer in an adequately fast and economically feasible manner, one must make use of existing technology which has already been used to resolve related problems in a larger application, as mentioned before. In this manner, it is suitable to make use of

passive UHF RFID technology. Since this technology does not make use of a battery, the participants are only required to be tagged and so no complications arise. Moreover, passive tags are low-priced, meaning that if a tag is somehow damaged during the race, a new tag can be simply collected and the participant can head back to the race.

5. EVALUATION PLAN

In order to setup and build the RFID system, a number of specialised hardware will be used. These include:

1. SparkFun Simultaneous RFID Reader - M6E Nano
2. UHF RFID Antenna (RP-TNC)
3. UHF RFID Tags
4. Computer System

The main component that is going to be utilised is the Simultaneous RFID Reader, which will allow us to be able to read the UHF RFID tags and gather the required data from them. The RFID Reader board is able to read up to 150 tags per second and has a writing time of tags at 80 milliseconds. It has an adjustable power output between 0dBm up to 27dBm, and allows for the use of the UHF RFID Antenna, which can read up to 4.9m. Different RFID tags will be used to tackle **O1**. This will ensure that the system is not limited to one single type of wearable device.

The software that will be employed to cover **O2** is the Universal Reader Assistant by ThingMagic. This will be used to implement a number of adjustable functions for the RFID reader, mainly:

1. Read and Write Options to manage reading time, filtering of tags and antenna detection
2. Performance Adjusting to fine tune the reading power, distance, and rate, and tag feedback
3. Performance Metrics to present the different unique tags and their reactions, as well as the read rate

A number of participants will then be required to be tagged with an RFID tag in different areas as discussed previously to test out the system. The results obtained from this testing will be used to determine the ideal placement and angle of the RFID antenna, as well as the optimal tagging of the RFID tags, while ensuring that **O2** is optimally covered. **O3** will be evaluated and experimented during the participants' testing.

6. CONCLUSION

This report puts forward the research that has been made so far. A number of reviews were made on RFID systems and how they were used in different sectors and environments for different purposes in order to establish the state of the art in this area. The proposed solution is put forward together with how it will be executed. Challenges such as RFID tag detecting and software issues are anticipated to arise during testing, which will optimistically be surmounted.

7. REFERENCES

- [1] S. M. Ali and A. Soleymani. Rfid-based object detection system in iot. 12 2016.
- [2] F. Aloul, A. Sagahyroon, A. Nahle, A. Dehn, and R. A. Anani. Guideme: An effective rfid-based traffic monitoring system, 2012.
- [3] M. Anu, M. Deepika, and L. Gladance. Animal identification and data management using rfid technology. pages 1–6, 11 2015.
- [4] L. Atzori, A. Iera, and G. Morabito. The internet of things: A survey. *Computer Networks*, 54:2787–2805, 10 2010.
- [5] B. Carbutar, M. K. Ramanathan, M. Koyutürk, S. Jagannathan, and A. Grama. Efficient tag detection in rfid systems. *Journal of Parallel and Distributed Computing*, 69:180–196, 2 2009.
- [6] D. P. Chechi, P. J. K. C. Bose, D. Parkash, T. Kundu, and P. Kaur. The rfid technology and its applications: A review, 2012.
- [7] L. Cui, Z. Zhang, N. Gao, Z. Meng, and Z. Li. Radio frequency identification and sensing techniques and their applications—a review of the state-of-the-art. *Sensors*, 19, 2019.
- [8] A. Czinkóczy and G. Szabó. Bicycle road optimization using rfid tags, 2010.
- [9] G. Deconinck and W. Grynczel. Open source rfid race timing system, 2017.
- [10] A. Ejder, G. Yilmaz, and K. Senem. The product-based tracking system by using rfid technologies in logistics applications. *The International Journal of Business Management*, 5, 11 2017.
- [11] A. El-Deeb, A. Foina, R. M. Badia, and J. Ramirez-Fernandez. Player tracker - a tool to analyze sport players using rfid. 11 2010.
- [12] D. F., V. Reza, and R. Alireza. Wearables and the internet of things (iot), applications, opportunities, and challenges: A survey. *IEEE Access*, PP:1–1, 04 2020.
- [13] C. Filippo, G. Simone, B. Michele, M. Andrea, D. Francesco, and M. Giuliano. A review of rfid sensors, the new frontier of internet of things. *Sensors (Basel, Switzerland)*, 21, 04 2021.
- [14] D. Iturralde, P. Peralta, A. Cabrera, and G. Delgado. Performance analysis of radio frequency identification technology in race timing systems, 11 2020.
- [15] S. Jayant. Automated classroom monitoring with iot and virtuino app. 8 no 3:1–17, 04 2019.
- [16] S. Joolrat, K. R. Saikaew, M. Tengrungrroj, P. Jiranuwattanawong, C. Janpanich, and K. Sornsakda. Rfid data processing in a real-time monitoring system for marathon. 43:94–99, 2015.
- [17] F. Kai, G. Yuanyuan, D. Zhao, L. Hui, and Y. Yintang. Rfid secure application revocation for iot in 5g. In *2015 IEEE Trustcom/BigDataSE/ISPA*, volume 1, pages 175–181, 2015.
- [18] A. Kamran, H. Shah, and K. Paul. Rfid applications: An introductory and exploratory study. *International Journal of Computer Science Issues*, 7, 02 2010.
- [19] A. Kemal, G. Ismail, A. Ramazan, P. Nezih, and K. Abdullah. Iot-based occupancy monitoring techniques for energy-efficient smart buildings. page

396, 2015.

- [20] F. Klaus. *RFID Handbook: Fundamentals and Applications in Contactless Smart Cards and Identification, Second Edition*, pages 61 – 159. 07 2003.
- [21] M. B. Kudri 'c, P. Šoli 'c, and N. Roži 'c. Rfid technology based object tracking system, 2013.
- [22] K.-Y. Lin, M.-W. Hsu, and S.-R. Liou. Bicycle management systems in anti-theft, certification, and race by using rfid. In *Proceedings of 2011 Cross Strait Quad-Regional Radio Science and Wireless Technology Conference*, volume 2, pages 1054–1057, 2011.
- [23] S. M. Masohor, M. Z. Bakar, W. S. Yamin, M. H. A. Wahab, M. S. Ahmad, A. A. Ma'Radzi, T. N. T. Ibrahim, and C. C. Choon. Tracking system using rfid for hiking activity with iot technology. volume 1529. Institute of Physics Publishing, 6 2020.
- [24] M. M.H., M. S. M.A., O. M.A., S. A., B. M.F., A. R. A., A. M.M.M., I. M.M., and Y. N. Smart le tour checkpoint using rfid. In *2014 International Symposium on Technology Management and Emerging Technologies*, pages 246–251, 2014.
- [25] P. Milan, S. Milan, and H. Premysl. The optimization of the rfid system for the identification of sportsmen in mass races. In *2007 European Microwave Conference*, pages 732–735, 2007.
- [26] P. G. G. Narkhede, A. K. Langade, K. Mehta, and S. A. Kulkarni. Rfid based tracking system. *IJARCCCE*, 6:1045–1049, 3 2017.
- [27] G. Pai, J. M. D'souza, J. T. D'souza, S. R. Acharya, and M. Chaitra. Health monitoring using rfid. *American Journal of Intelligent Systems*, 2017:100–103, 2017.
- [28] S. Pawut, J. Rachsuda, K. S. Won, and S. Pisal. A cost-effective interactive platform for the management of a small scale lap-based jogging competition using low-frequency rfid technology. In *2017 3rd International Conference on Science in Information Technology (ICSITech)*, pages 360–365, 2017.
- [29] S. Petar, B. Zoran, S. Maja, P. Luigi, C. Riccardo, Rodrigues, and J. J. P. C. Gen2 rfid as iot enabler: Characterization and performance improvement. *IEEE Wireless Communications*, 24(3):33–39, 2017.
- [30] S. U. Rehman, R. Liu, H. Zhang, G. Liang, Y. Fu, and A. Qayoom. Localization of moving objects based on rfid tag array and laser ranging information. *Electronics (Switzerland)*, 8, 8 2019.
- [31] W. Roy. The magic of rfid, 2004.
- [32] M. F. Saaïd and M. N. Handani. Rfid-based cashless payment system. 2014.
- [33] S. Shubham, M. Suvojit, and D. Subhadeep. Smart bag tracking and alert system using rfid. In *2017 International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT)*, pages 1–4, 2017.
- [34] L. Tan and N. Wang. Future internet: The internet of things. In *2010 3rd International Conference on Advanced Computer Theory and Engineering (ICACTE)*, volume 5, pages V5–376–V5–380, 2010.
- [35] M. Tengrungrroj, P. Jiranuwattanawong, S. Joolrat, D. Srisumran, K. Tientanopajai, A. Chaosakul, and K. R. Saikaew. Performance improvement of rfid system in mass races, 2014.
- [36] W.-F. Tsai, R. J. Burkholder, and F. L. Teixeira. A low cost rfid tracking and timing system for bike races, 2011.
- [37] M. J. Uddin, M. I. Ibrahimy, M. B. I. Reaz, and A. N. Nordin. Design and application of radio frequency identification systems, 2009.
- [38] A. Ullah. Iot: Applications of rfid and issues. *International Journal of Internet of Things and Web Services*, 3, 2014.
- [39] S. F. Wamba and E. W. Ngai. Importance of the relative advantage of rfid as enabler of asset management in the healthcare: Results from a delphi study. pages 2879–2889. IEEE Computer Society, 2012.
- [40] Y. Wen, C. Chao-Hsien, and L. Zang. The use of rfid in healthcare: Benefits and barriers, 2010.
- [41] V. K. Y. Wu, N. H. Vaidya, and R. H. Campbell. Rfid trees: A distributed rfid tag storage infrastructure to backtrack hikers in a forest *, 2009.
- [42] J. Xiaolin, F. Quanyuan, F. Taihua, and L. Quanshui. Rfid technology and its applications in internet of things (iot), 2012.
- [43] A. Yang and G. P. Hancke. Rfid and contactless technology, 5 2017.
- [44] B. S. Çiftler, A. Kadri, and I. Güvenç. Iot localization for bistatic passive uhf rfid systems with 3-d radiation pattern. *IEEE Internet of Things Journal*, 4:905–916, 8 2017.